

1 Background and Notations

1.1 Poisson Processes

Definition 1. A counting process $N = (N(t))_{t \geq 0}$ is a stochastic process such that :

- $N(0) = 0$,
- $N(t)$ takes values in the set of non-negative integers,
- $N(t)$ is non-decreasing and right-continuous.

Definition 2. A Poisson process with rate $\lambda > 0$ is a counting process $N = (N(t))_{t \geq 0}$ such that :

- $N(0) = 0$,
- The process has independent increments,
- The number of events in any interval of length t follows a Poisson distribution with parameter λt .

1.2 Conditional Poisson Processes

Definition 3. A conditional Poisson process is a counting process $N = (N(t))_{t \geq 0}$ such that, given a stochastic intensity function $\lambda(t)$, the process behaves like a Poisson process with rate $\lambda(t)$ at time t .

1.3 Cramér-Lundberg Model

As an insurance company, the main goal is to accurately model the risk associated with claim occurrences and sizes. Doing so enables the company to set appropriate premiums, maintain sufficient reserves, and ensure long-term profitability, i.e., a growing surplus.

We formally define the surplus as follows :

$$R(t) = u + ct - S(t), \quad (1)$$

with u the initial reserve, c the premium rate, and $S(t)$ the aggregate claims up to time t .

In order to model $S(t)$, we consider the classical Cramér-Lundberg model which assumes that :

$$S(t) = \sum_{i=1}^{N(t)} X_i, \quad (2)$$

where $N(t)$ is a Poisson process with rate $\lambda > 0$, representing the number of claims up to time t . The X_i are i.i.d. random variables representing the claim sizes, **independent** of $N(t)$, with common distribution F_X .